IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

ELECTROMAGNETISM

Wednesday, 7 May 2025, 2:30 PM

Time allowed: 2:00 hours

There are THREE questions on this paper.

Answer ALL questions.

Any special instructions for invigilators and information for candidates are on page 1.

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Formula sheet

Universal constants: $\varepsilon_0 = 8.84 \times 10^{-12}$ F/m $\quad \mu_0 = 4\pi \times 10^{-7}$ F/m $\quad c = 3 \times 10^8$ m/s

Maxwell's equations in integral form

$$\oiint_S \mathbf{D} \cdot \mathrm{d}\mathbf{s} = \iiint_V \rho \mathrm{d}v$$
$$\oiint_S \mathbf{B} \cdot \mathrm{d}\mathbf{s} = 0$$
$$\oint_L \mathbf{H} \cdot \mathrm{d}\mathbf{l} = \iint_S \mathbf{J} \cdot \mathrm{d}\mathbf{s} + \frac{\mathrm{d}}{\mathrm{d}t} \iint_S \mathbf{D} \cdot \mathrm{d}\mathbf{s}$$
$$\oint_L \mathbf{E} \cdot \mathrm{d}\mathbf{l} = -\frac{\mathrm{d}}{\mathrm{d}t} \iint_S \mathbf{B} \cdot \mathrm{d}\mathbf{s}$$

Maxwell's equations in differential form

$$\nabla \cdot \mathbf{D} = \rho$$
$$\nabla \cdot \mathbf{B} = 0$$
$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$
$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

Constitutive relations: $\mathbf{D} = \varepsilon_0 \varepsilon_\mathrm{d} \mathbf{E} \qquad \mathbf{B} = \mu_0 \mu_\mathrm{r} \mathbf{H} \qquad \mathbf{J} = \sigma \mathbf{E}$

Boundary conditions

$$(\mathbf{D}_1 - \mathbf{D}_2) \cdot \mathbf{n} = \sigma \qquad (\mathbf{B}_1 - \mathbf{B}_2) \cdot \mathbf{n} = 0$$
$$\mathbf{n} \times [\mathbf{E}_1 - \mathbf{E}_2] = 0 \qquad \mathbf{n} \times [\mathbf{H}_1 - \mathbf{H}_2] = \mathbf{J}_\mathrm{s}$$

Solution of Poisson's equation for electrostatic potential

$$\varphi(\mathbf{r}) = \frac{1}{4\pi\varepsilon_0} \iiint_V \frac{\rho(\mathbf{r}')\mathrm{d}v}{|\mathbf{r} - \mathbf{r}'|}$$

Biot-Savart law

$$\mathbf{B} = \frac{\mu_0}{4\pi} I \int_L \frac{\mathrm{d}\mathbf{l} \times \mathbf{r}}{r^3}$$

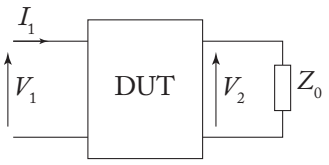
Transmission-line equations

$$\frac{\partial I(z,t)}{\partial z} = -C \frac{\partial V(z,t)}{\partial t} \qquad \text{and} \qquad \frac{\partial V(z,t)}{\partial z} = -L \frac{\partial I(z,t)}{\partial t}$$

Input impedance of a loaded transmission line

$$Z_{\text{in}} = Z_0 \frac{Z_{\text{L}} + jZ_0 \tan(kd)}{Z_0 + jZ_{\text{L}} \tan(kd)}$$

S-parameters for a two-port device



$$S_{11} = \frac{V_1 - Z_0 I_1}{V_1 + Z_0 I_1} \quad \text{and} \quad S_{21} = \frac{2V_2}{V_1 + Z_0 I_1}$$

Fields of a short dipole antenna (far-field region)

$$E_\theta = jZ_0 \frac{I_0 kd}{4\pi} \frac{e^{-jkr}}{r} \sin\theta$$
$$H_\varphi = \frac{E_\theta}{Z_0}$$

Fields of a half-wave dipole antenna (far-field region)

$$E_\theta = \frac{jZ_0 I_0}{2\pi} \frac{\cos\dfrac{\pi\cos\theta}{2}}{\sin\theta} \frac{e^{-jkr}}{r}$$
$$H_\varphi = \frac{E_\theta}{Z_0}$$

Gauss' and Stokes' theorems

$$\oiint_S \mathbf{F} \cdot \mathrm{d}\mathbf{s} = \iiint_V \nabla \cdot \mathbf{F}\, \mathrm{d}v \qquad \oint_L \mathbf{F} \cdot \mathrm{d}\mathbf{l} = \iint_S [\nabla \times \mathbf{F}] \cdot \mathrm{d}\mathbf{s}$$

Useful integrals

$$\int \frac{\mathrm{d}x}{\sqrt{x^2 + a^2}} = \ln(x + \sqrt{a^2 + x^2}) \quad \int \frac{\mathrm{d}x}{(x^2 + a^2)^{3/2}} = \frac{x}{a^2\sqrt{x^2 + a^2}} \quad \int \frac{x\,\mathrm{d}x}{(x^2 + a^2)^{3/2}} = -\frac{1}{\sqrt{x^2 + a^2}}$$

ELECTROMAGNETISM

1.

a) Write down the differential form of Maxwell's equations in free space and for harmonic time variation of all quantities in the form $e^{j\omega t}$. [5]

b) Show that $\mathbf{E} = \sin(x)\cos(y)\,\mathbf{e}_x + 0\,\mathbf{e}_y + 0\,\mathbf{e}_z$ *cannot* be a valid expression for the field strength of a static electric field. [5]

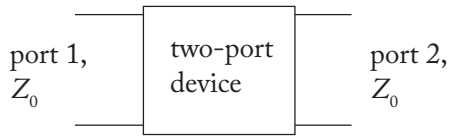
c) Consider a static magnetic field whose strength is given by

$$\mathbf{H} = z\,\mathbf{e}_x + x\,\mathbf{e}_y + y\,\mathbf{e}_z$$

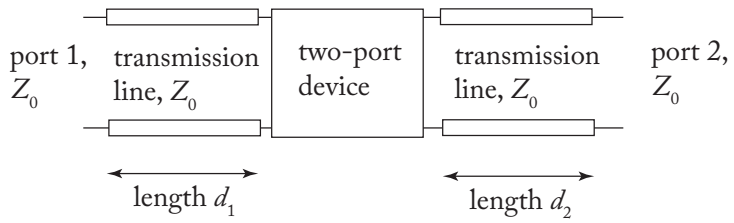
Show that this expression satisfies Maxwell's equations and derive an expression for the current density. Assume $\mathbf{B} = \mu_0\mathbf{H}$. [10]

d) Using a general expression from the formula sheet, find an expression for the input impedance of a transmission line terminated by an impedance Z_L when the length of the transmission line is equal to a quarter wavelength at the frequency of interest. The characteristic impedance of the transmission line is Z_0. [5]

e) The values of two scattering parameters of the two-port device shown schematically in Fig. 1.1(a) were measured as S_{11} and S_{21} for incidence from port 1. The port impedances were Z_0. The device was then connected to two sections of transmission lines with lengths d_1 and d_2, as shown in Fig. 1.1(b). The characteristic impedances of both transmission lines were equal to Z_0 (same as the port impedances). Find the values of the scattering parameters measured for the system of Fig. 1.1(b) when connected to the same ports. [10]



(a)



(b)

Figure 1.1

f) Write down a relationship between the angular frequency ω and the wavenumber k of a plane transverse electromagnetic wave in free space. Explain the meaning of any additional parameters in this relationship. [5]

2. a) Explain, using derivations as necessary, how a vector network analyser can be used as an impedance meter. [15]

b) Suggest one advantage and one disadvantage of using a vector network analyser for impedance measurements. Justify your answers. [5]

c) Find the S-parameters of the device shown in Fig. 2.1. [10]

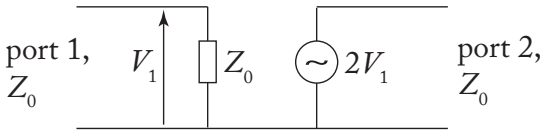


Figure 2.1

3. The vector wave equation for a metal with a conductivity σ can be written in the form

$$\nabla^2\mathbf{E} + \omega^2\mu_0\varepsilon_0\left(1 - j\frac{\sigma}{\omega\varepsilon_0}\right)\mathbf{E} = 0$$

assuming harmonic variation with an angular frequency ω.

a) Write a scalar form of the wave equation assuming a z-going, x-polarised plane electromagnetic wave. [5].

b) For this wave, derive a relationship between the angular frequency and the wavenumber. [10]

c) Derive an approximate expression for the wavenumber assuming that the conductivity is high. [5]

d) Define skin depth and derive an expression for skin depth in the high-conductivity approximation. [10]

ELEC50007 Electromagnetism final exam: errors

Candidates failed to:

Question 1
a) state that ρ=0, J=0 (common); state relationship between E, H and B, D (common)
b) evaluate curl E correctly (rare)
c) evaluate curl H, evaluate div B (rare)
d) evaluate the expression with infinities in numerator and denominator (rare)
e) approach the problem correctly (very common). Most candidates failed to notice that the effect of the transmission lines amounts to additional phase shift of S-parameters. Some candidates stated the S-parameters did not change, which applies only to their absolute values.

Question 2
a) appreciate that the concept of impedance applies to one-port devices (common)
b) offer a reasonable disadvantage (common). Options often offered apply to any measurement method (such as noise, need for calibration, influence of temperature).
c) evaluate the S-parameters correctly, most often S_{12} and S_{22} (common); evaluate all four S-parameters (common)

Question 3
a) derive ordinary differential equation (common)
b) derive correct relationship between the angular frequency and the wavenumber (common)
c) derive the complete relationship (common)
d) define skin depth correctly; derive correct expression for skin depth (common)